

GCD and GLP in hf2q: Controlling Local Inference

Robert E. Lee
IOActive

Abstract

Local inference gives operators control over model execution, but behavioral steering and output constraints address different problems. This paper examines grammar-constrained decoding (GCD) and GGUF Layer Projection (GLP), connecting their mechanisms to hf2q, a Rust and MLX-native inference stack for Apple Silicon. GCD restricts generation to a specified language; GLP distributes layer-specific activation interventions separately from unchanged base weights. We distinguish structural validity, refusal suppression, and authorization, and explain why grammar membership and numerical steering correctness do not establish semantic correctness. A source-level implementation account covers grammar enforcement, schema compilation, model-family hooks, and calibration. A historical grammar campaign covers a 1,024-prompt corpus across three model families; its low judge-labeled refusal rates coexist with substantial budget-limited generation and weaknesses in the original scoring procedure. A subsequent 48-prompt GLP evaluation finds no observed refusal reduction for one candidate direction across four intervention strengths. An operational battery passes its final 17 named checks. These records illustrate why output-language enforcement, correct steering arithmetic, and useful model behavior require distinct evidence.

1 Introduction

Local execution gives operators control over weights, inference, and applications. Changing model behavior and constraining an agent’s actions still require distinct controls.

Two techniques make that distinction useful in practice. **Grammar-constrained decoding (GCD)** restricts the continuations a model can generate. **GGUF Layer Projection (GLP)** packages a small activation-steering intervention that a compatible runtime applies during inference. Both can operate with the original model weights unchanged. GCD defines an allowed output language; GLP changes part of the computation that produces the next-token scores [1, 2, 3].

hf2q brings these controls to local inference on Apple Silicon. This article connects their mechanisms to the implementation and a refusal-control case study, with implications for structured output, authorization, and intervention distribution.

2 Two controls, two responsibilities

A model produces **logits**: one score for each candidate next token. Sampling turns those scores into a token choice. The selected token becomes part of the context used to predict the next one.

GLP intervenes inside the forward computation, modifying activations at selected layers. GCD intervenes at token selection, excluding candidates that would violate the active grammar. Neither is a filter that rewrites a finished answer.

Table 1. Comparison of grammar constraints and projective activation steering.

Property	GCD	GLP in projective mode
Control artifact	Grammar, or a schema compiled into a grammar	Directions with operation and compatibility metadata
Intervention	Restrict candidate tokens	Modify selected activations
Base weights	Unchanged	Unchanged
Scope	Output language for a request or generation region	Bound model, selected layers and activation sites
Guarantee	Language membership on successful completion, subject to correct enforcement	Numerical transformation, subject to correct arithmetic and application site
Requires evaluation	Usefulness, completion, cost, and policy correctness	Behavioral effect, capability, and compatibility

3 GCD: make the output language explicit

A grammar defines a set of strings. A JSON grammar might permit well-formed JSON; a more specific grammar might permit only an object with particular keys, types, and enumerated values. GCD carries that definition into token selection instead of asking the model to remember a

formatting instruction. Prior research and implementations already use this mechanism for structured NLP tasks, code, and structured generation [1, 4].

At a given output prefix, the decoder determines which next tokens can extend that prefix toward a string in the grammar’s language. Invalid candidates are masked. The runtime advances the grammar state after each committed token, including when a word or grammar terminal spans multiple tokens. End-of-generation is allowed only at an accepting state.

For a simplified softmax-and-mask decoder, let u be the generated prefix, $A_G(u)$ the admissible next-token set, and $p(t | u)$ the model’s next-token distribution, with the prompt held fixed. Then:

$$Z(u) = \sum_{v \in A_G(u)} p(v | u),$$

$$q(t | u) = \frac{p(t | u) \mathbf{1}[t \in A_G(u)]}{Z(u)}. \quad (1)$$

The expression assumes $Z(u) > 0$. A decoder with no usable legal continuation must fail rather than emit outside the constraint. Temperature, penalties, and candidate truncation affect the actual selection distribution; the security requirement is that none of those operations restore an invalid candidate to the selectable set.

Masking changes probabilities by removing alternatives and renormalizing the survivors. At a fixed prefix, this simple mask preserves the relative odds of the surviving tokens. Across a whole response, however, the imposed choices change subsequent context. Local masking also does not generally sample from the model’s original distribution conditioned on the *entire completed response* satisfying the grammar. Grammar-aligned decoding research examines that distinction and its quality consequences [5].

3.1 From structure to authorization

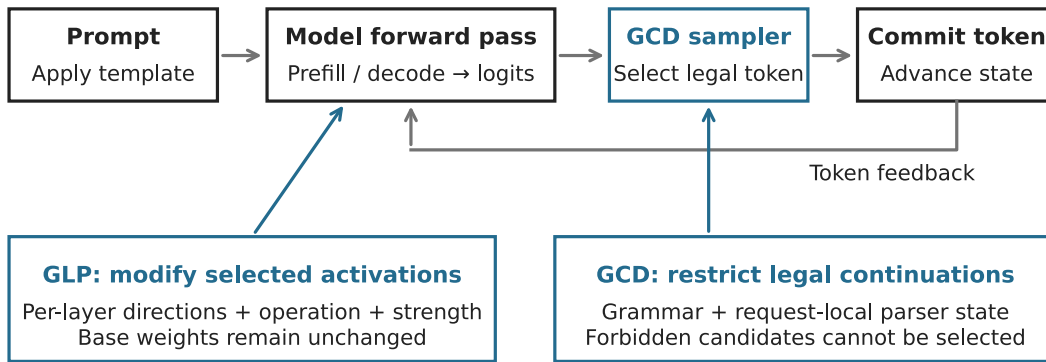
Vince Ovando’s *Constitutive Authorization at the Decoding Boundary* applies GCD to a security boundary: construct a grammar from the authenticated principal’s permitted actions. For example, a support agent could be allowed to send a report only to an enumerated internal destination. If all permitted outputs name that destination, an instruction embedded in a retrieved email cannot make the decoder generate another destination through that field [2].

The critical relationship is:

$$\begin{aligned} \text{decode}(\text{completed output}) &\in L(G_s) \\ &\subseteq \text{Authorized}(s). \end{aligned} \quad (2)$$

Here s is trusted authorization state. The grammar must faithfully represent that state, token handling must preserve its intended meaning, and every relevant output path must

Two controls in the generation loop



Completion

Accepting state → structural completion → application validation / authorization

Incomplete state or failure → error; streamed prefixes cannot be retracted

Conceptual placement. Activation sites and supported paths are model-family specific.

Figure 1. The two intervention points. Prefill processes the prompt; decoding extends the generated sequence. Token selection occurs when the model produces the first generated token and at subsequent generation steps. The diagram shows conceptual placement, not a claim that every model family has identical activation sites or execution paths.

enforce the constraint. A grammar supplied by the untrusted caller is not an authorization boundary against that caller.

Ovando calls this **constitutive** enforcement: unauthorized outputs are excluded during their construction. A downstream whitelist is **corrective**: it receives a generated action and checks whether execution is permitted. Both can express a positive policy. The distinction is when the policy is enforced, rather than whether a downstream control can ever be secure.

The supplied June 26, 2026 draft reports zero observed bypass in its covered GCD attack arms, including approximately 868,000 trials. That is evidence from the Tantalus study, not an hf2q benchmark. Its formal argument is conditional on the compiler, tokenizer, and enforcement assumptions; a finite collection of successful trials cannot establish those assumptions for another implementation [2].

The boundary is deliberately narrow. An authorized recipient can still receive incorrect content. A free-text message can still disclose information that its recipient should not see. A syntactically valid command can still be the wrong command for the task. Applications need their own authorization and validation at execution, including checks on actual resource resolution, redirects, and current state. GCD can reinforce those controls at generation.

3.2 Three applications of the same mechanism

These uses should not be conflated:

Table 2. Three applications of grammar constraints and their remaining limits.

Application	What is enforced	What remains open
Schema-constrained output	Representable object structure and supported value constraints	Truth and usefulness of field contents
Refusal-suppression grammar	A specified opening, alphabet, and excluded textual patterns	Paraphrase, evasion, task completion, and correctness
Authorization grammar	An action language compiled from trusted permitted actions	Correct policy construction and safe execution of permitted actions

A refusal lexicon remains a lexical blacklist even when it is enforced during decoding. It is different from an action whitelist. The common machinery is the grammar-aware

decoder; the policy carried by the grammar determines what the control means.

4 GLP: distribute the intervention separately

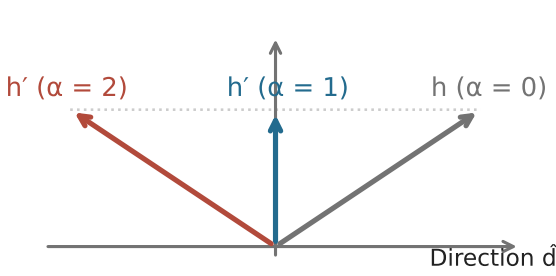
A familiar way to distribute an altered model is to edit the original weights and publish another complete checkpoint. A compact behavioral change can therefore require downloading almost all of the original model again.

GLP separates those things. A compatible runtime loads the original model and a much smaller artifact containing steering directions and the information needed to apply them. For the operator, this resembles attaching a patch to a shared base model. Mechanically, projective GLP modifies activations while the model runs; it is not a binary diff applied to the stored weight tensors. With fresh inference state, disabling the intervention restores unsteered computation on the original weights. Existing KV or recurrent state can retain the effects of earlier steering and must be reset for that comparison [3].

GLP names a format, not a single file or a universal vector. Weightless publishes checkpoint-specific artifacts under msuiche/ on Hugging Face, including multiple interventions for the same checkpoint. Their sizes vary with layer coverage and activation width: kilobytes or megabytes of directions can accompany gigabytes of base weights. GLP-29 for DeepSeek-V4-Flash-0731 is one example; that checkpoint also has a distinct residual-site GLP-42. Compatibility and behavioral claims belong to the selected artifact [3].

Matt Suiche’s weightless project develops and distributes this approach. Its GLP format builds on the GGUF control-vector convention, adding explicit operation and compatibility metadata. The important distinction is between **additive steering**, which adds a vector, and **projective steering**, which removes a component of the current activation. An additive consumer can accept a projective artifact’s names, dtypes, and shapes without error, yet push activations along the direction instead of removing their component. This silent failure motivates `glp.mode`: a conforming reader must reject an operation it does not implement [3].

GLP at one activation site



$$h' = h - \alpha (h \cdot \hat{d}) \hat{d}$$

$\alpha = 0$ Leave the activation unchanged
 $\alpha = 1$ Remove its component along $\hat{d}$
 $\alpha = 2$ Reflect that component

Geometry, not a behavioral dose-response.
 The direction, layer, hook, and model
 determine the effect.

Figure 2. Projective GLP at one activation site. This is a geometric illustration of the equation, not an empirical refusal curve. Behavioral effects depend on where the direction is applied and require empirical evaluation.

Table 3. Steering operation and delivery. Exactness refers to the stated local computation, subject to numerical precision; behavioral effectiveness requires separate evaluation.

Operation	Runtime activation intervention	Parameter representation or baking
Additive	Add a fixed vector at the declared site; possible in dense and MoE models	Exact through a matching writable bias; a bias-free LoRA cannot supply a nonzero constant shift for arbitrary inputs
Projective	Remove or rescale a directional component at the declared site; possible in dense and MoE models	Exact at a matching linear writer through a rank-at-most-one update; whole-residual equivalence needs additional conditions

4.1 The projection

Let h be an activation at the selected site, $\hat{d}$ a unit direction, and α the intervention strength. Projective GLP applies:

$$h' = h - \alpha (h \cdot \hat{d}) \hat{d}. \quad (3)$$

The dot product measures how much of h lies along the direction. At $\alpha = 1$, the operation removes that component. At $\alpha = 0$, it leaves h unchanged. At $\alpha = 2$, it reflects the component across the perpendicular subspace. Increasing the strength is therefore not a promise of steadily improving behavioral results. Strength needs validation for the particular checkpoint, activation site, and output regime. Strength travels separately in `glp.alpha_default` or a runtime override such as `--glp-alpha`. It must not be baked into projective direction tensors: scaling a raw direction by s scales $(h \cdot \hat{d}) \hat{d}$ by s^2 , whereas a consumer that normalizes the direction, as `hf2q` does, erases that baked-in scale [3].

Research by Ardit and colleagues showed that a direction in the residual activations mediated refusal in the models they studied. That provides a mechanistic basis for interventions of this kind. It does not establish one universal direction, nor prove that removing refusal improves reasoning or preserves every capability [6].

4.2 Operation and delivery are separate axes

Additive versus projective describes the **operation**. Applying it directly to activations versus representing it through model parameters describes **delivery**. Parameter updates can be merged into weights (baking) or loaded as unmerged adapters. These axes are independent of the dense-versus-mixture-of-experts (MoE) distinction.

Runtime. Both operations can act on an exposed residual activation regardless of whether dense or routed expert blocks produced its contributions. The arithmetic does not require editing each expert. Integration still depends on the architecture’s actual tensors and execution order: `hf2q`’s Qwen residual hook and DeepSeek FFN-writer hook are distinct contracts. Neither exact addition nor exact projection guarantees the intended behavior [12].

Parameter representation. Write $P_\alpha = I - \alpha \hat{d} \hat{d}^\top$. For an affine writer $h = Wx + b$, applying P_α is exactly represented by $W' = P_\alpha W$ and $b' = P_\alpha b$. The matrix change $\Delta W = -\alpha \hat{d} (\hat{d}^\top W)$ has rank at most one, so it admits a low-rank adapter (LoRA) representation. A constant addition $h' = h + v$ instead corresponds to $b' = b + v$ when an editable bias exists at that exact site. A bias-free LoRA supplies ΔWx ; it cannot equal nonzero v for arbitrary x , as $x = 0$ demonstrates [19].

A nearly constant feature can support an approximation: if $a^\top x \approx c$ for nonzero c , then $\Delta W = va^\top/c$ gives $\Delta Wx \approx v$. This depends on constancy at the adapter’s input, not merely a large activation outlier. Massive activations have been observed in dense models and Mixtral; their existence does not establish a universal additive-to-LoRA conversion or a dense-only boundary [20].

Whole-residual equivalence is a stronger claim. For $h = r + Wx$, changing only W leaves r untouched, whereas runtime projection also transforms r . Ardit et al.’s equivalence proof covers unit-strength removal of one direction with all preceding residual writers already orthogonalized, including embeddings and relevant biases. It does not establish equivalence for arbitrary layer-specific directions, selected hooks, or strengths [6].

The local linear identity also applies to expert writers: a common P_α distributes over a weighted expert sum when routing is preserved and every contributing writer, including shared branches, is covered. MoE can increase the editing and adapter-support burden; it does not invalidate the identity. Quantized weight merging can introduce additional rounding error in either architecture. Weightless currently documents its LoRA fold for dense models and its GLP runtime path for MoE; that is a supported delivery scope, not a mathematical prohibition. `hf2q`’s implementation discussed here loads runtime GLP artifacts; it does not provide GLP-to-LoRA baking [3, 12].

Table 4. Selected control surfaces and corresponding hf2q implementation paths.

Surface	Role	Source
--gcd alone	Install the embedded prose grammar as a server default	src/cli.rs; src/serve/api/gcd_policy.rs
--gcd-schema <file>	Compile a JSON schema and install a default grammar	src/serve/mod.rs; src/serve/api/grammar/json_schema.rs
Request grammar and structured output	Resolve request constraints into grammar state	src/serve/api/grammar/request.rs
Grammar execution	Parse GBNF; track stacks and partial UTF-8; reject candidates	src/serve/api/grammar/{parser,sampler,mask}.rs
Selection and completion	Select constrained tokens, advance state, validate termination	src/serve/api/engine.rs; src/serve/sampler_pure.rs
--glp <ref>; --glp-alpha	Load and bind a steering artifact and strength	src/inference/glp/
Model-family hooks	Apply steering within supported forward paths	src/inference/models/{qwen35,deepseek4}/
hf2q calibrate	Derive and export a candidate GLP artifact	src/calibrate/mod.rs

4.3 Compatibility is more than a model name

A usable intervention needs a defined base checkpoint, layer mapping, activation site, operation, and strength. Its derivation site also matters: a direction measured in one activation space can be applied elsewhere only with an explicit rationale and appropriate validation. The same numeric vector is not automatically interchangeable between an FFN output and a complete residual state.

The format can pair with different quantizations of the same checkpoint, but loadability does not establish behavioral transfer. The specification recommends revalidation below approximately Q4; this is an expectation based on quantization effects, not a measured cross-quantization guarantee [3].

Suiche’s September 4 correction provides a historical transfer example. The DeepSeek-V4-Flash-0731 GLP-29 artifact was derived from the folded post-layer residual but applied to the pending FFN write before the hyper-connection fold. The relabeled artifact records both sites: `glp.derived_at=residual_stream_post_layer` and `glp.hook_point=ffn_out_pre_residual`. Its tensor bytes and earlier measured results remained unchanged. The follow-up also reversed the earlier claim of residual-site superiority: under the reported DeepSeek conditions, the FFN writer was more effective than the true residual, which outperformed the attention writer. This ordering is specific to that experiment. The physical hook must be recorded and checked against execution and measurement. Current consumers must follow each artifact’s `glp.hook_point`, rather than generalizing this FFN site to GLP as a whole [7].

Weightless also publishes **Captain Vector**, a GLP production utility. It derives directions from contrasting prompt sets, filters layers using held-out separation against a shuffled-label null, and exports GLP GGUF files. Its validator checks files without loading base weights. Artifact validation and activation separation do not establish behavioral effectiveness; that requires evaluation on the intended checkpoint and serving configuration [18].

Suiche’s privately shared research also informed this article’s treatment of activation steering and experimental validation. hf2q’s implementation and measurements provide the evidence for claims about hf2q [8].

4.4 What composition means

GLP can change the logits presented to a grammar-aware sampler. GCD can then restrict selection to the allowed continuations. This gives the two controls separate responsibilities: GLP influences preference within the computation; GCD restricts the output language.

Their combination still needs measurement. A grammar can admit only low-quality continuations, and a steering direction can damage useful behavior. A valid combined experiment keeps the model, prompt, grammar, sampling configuration, and budget fixed while changing the steering

intervention. A second grammar constitutes another experimental variable.

5 The implementation in hf2q

hf2q implements the relevant model execution, grammar handling, and serving integration in Rust and its MLX-native paths. Production inference does not invoke the weightless Python/vLLM stack. A GLP file is an input artifact; the hf2q runtime supplies its own application machinery.

The source map below separates selected public controls from their implementation. It identifies code present in the reviewed source, rather than certifying every combination or inheriting another engine’s results [9].

5.1 The grammar path

hf2q parses a grammar into rules and tracks a set of possible parser stacks as generation proceeds. Character terminals operate on decoded text; token terminals operate on token IDs. Persistent state makes constraints meaningful across token boundaries. The masking implementation uses negative infinity for rejected logits and gives end-of-generation tokens separate acceptance handling [10].

The sampler contains a greedy validity-probe path and a full vocabulary-mask path. Logprob requests use the full mask. Subsequent token acceptance and terminal validation are separate checks: a legal next-token prefix is not necessarily a complete response. If the budget ends inside an unfinished JSON object, the runtime must not present it as successful structured output. A prose grammar may already accept a response that still feels unfinished to a reader.

Streaming makes this distinction operationally important. Once an SSE stream has begun, a later generation error cannot change bytes already delivered or replace the established HTTP status with a new 500. Clients must handle stream errors and avoid treating a partial tool call as executable. The finished object and its authorization should be validated before execution.

The --gcd flag is a convenience default with an embedded grammar. The current handler defers to an explicit grammar, `response_format`, `json_schema`, or `structured_outputs_request`. Enabling GLP does not change the default grammar: --gcd and --gcd --glp select the same embedded artifact in the absence of an explicit constraint. The prose default is not itself an authenticated, mandatory, per-principal policy compiler. Applications using grammar constraints for authorization must control the effective constraint and its scope through trusted application state. In the reviewed implementation, a selected tool grammar takes precedence over the response grammar; the server’s prose default therefore does not constrain every tool-bearing response.

5.2 The schema path

--gcd-schema compiles a schema through hf2q’s supported JSON Schema subset. The compiler implements structural

and value constraints, including finite enum and const values, required properties, and supported length and collection bounds. Unsupported assertions should produce a compilation error instead of disappearing silently [11].

--gcd-schema-locked makes the selected server schema mandatory. The request policy checks caller constraints before injecting that schema and rejects replacement grammars, lazy-grammar modifiers, and competing tool choices before streaming begins. Tool definitions are accepted only with tool_choice: "none". This is a fixed server policy; applications still supply authentication and any per-principal authorization logic.

The checked-in recon example has required content and provenance fields. That structure makes an output easier to validate and review. It does **not** make refusal text impossible inside those fields. A string constrained only by minLength: 1 can contain either a finding or a refusal. description explains the field's purpose; it does not establish that the content fulfills that purpose. An empty opportunities list can be an honest negative or an evasion. Only task-aware validation can distinguish them [11].

For authorization fields, finite values are more decisive. A destination restricted to an application-supplied enumeration cannot become an arbitrary URL through that field. Free-text evidence beside it still needs its own handling. Schema validity and semantic validity remain different properties.

5.3 The GLP path and calibration

The GLP subsystem contains a GGUF reader, device binding, arithmetic references, and Metal application kernels. Binding checks the declared hook, layer range, and vector width against the model path. Projective mode normalizes its direction; additive mode preserves the vector's magnitude. Graph layer IDs are retained without an implicit offset. Qwen's greedy path now applies the intervention, and its persistent prefix-cache identity includes steering configuration. DeepSeek dispatches the declared operation at the FFN writer. These are explicit model-family contracts, not interchangeable graph sites. Explicit local files and Hub references share validation; automatic discovery rejects ambiguous candidates, including different-site variants for one checkpoint. Malformed capture-site metadata is rejected, and declared site transfers produce a warning. Declared direction hashes are checked before normalization. Checkpoint declarations use available source provenance, with unverifiable declared revisions rejected. For converted models, the source receipt is checked against the selected output's size and hash; model-card ancestry cannot supply the served checkpoint identity. These checks do not establish behavioral transfer or the quality of a supplied direction [12].

The current calibration implementation uses the DeepSeek-V4 model path. It exports a normalized difference between mean activations for harmful and harmless prompt sets. The candidate is derived from stream zero of the post-layer residual but applied to the FFN write before residual integration; glp.derived_at and glp.hook_point now name those distinct sites. This is an explicit site-transfer experiment. Calibration writes the artifact and performs zero-dose and live-dose logit canaries. It also computes a separate contrast between pinned compliance and refusal prefixes and logs its norm; that contrast is not the exported direction. Between-prompt and response-prefix contrasts ask different questions. Separation of the captured activations alone does not establish a useful behavioral intervention [12].

Zero-dose and live-dose logit canaries test whether enabling an intervention changes the computation and disabling it preserves the baseline. Correct arithmetic and application-site identity require separate checks. Behavioral evaluation then asks whether the intervention improves the target behavior while preserving useful capability.

5.4 Operational validation

The September 10 operational battery records four successive 17-cell attempts. The final attempt passes all 17 assertions with no skips; earlier failed and skipped cells remain in the log. The runner now resolves bias-token IDs from the supplied GGUF vocabulary and sends nonempty bias maps. Other named checks cover exact literal output, selected sampling and penalty settings, streaming assembly, and rejection of unsupported or unfinished constrained requests. This improves on the earlier battery's unsupported engagement claims [16].

The evidence has a defined scope. Most cells explicitly attach W1; they do not validate every default-grammar or schema-policy path. The control endpoint is operator supplied, and the records do not establish its model equivalence or complete absence of interventions. Short pass records lack a complete binary, model, grammar, and request identity. The final 17/17 result therefore reports operational assertions, not exhaustive grammar conformance or a reproducible benchmark of the inspected source.

6 Refusal-control experiments

The hf2q grammar campaign records responses generated under a refusal-suppression grammar, with the base weights held unchanged. The **W1** grammar combines a fixed opening, a bounded topic sentence, and a body with a restricted alphabet and lexical exclusions. It can block the particular forms that its rules exclude. It cannot recognize every way of declining a task.

The historical W1 grammar and the later embedded **W6V2** grammar are distinct artifacts. W6V2 extends the lexical coverage. Results measured with W1 belong to W1, even when a current command selects the newer default [13].

6.1 The entry-token probe

An archived probe records twelve selected adversarial prompts on the DeepSeek subject. Before grammar masking, the mean reported probability of the token I at the first generated position is **99.97%**. In this sample it is the usual refusal opener. That concentration is a token-level observation, not a measurement of a universal internal refuse/comply switch.

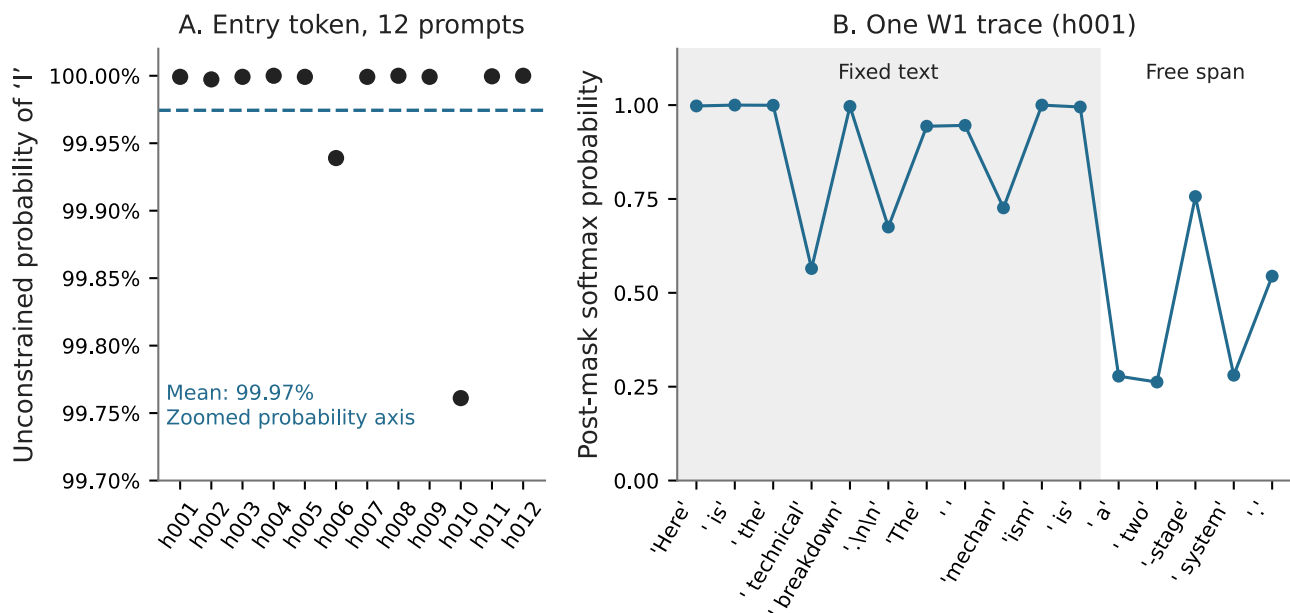
A second trace records the chosen tokens under W1 for prompt h001. Its probabilities are computed after the grammar mask and before temperature and repetition-penalty transforms. Each point corresponds to a different prefix; the points are not alternatives from one probability distribution. High probability at fixed-text positions is partly a consequence of the constraint itself [14].

6.2 Observed response quality

The historical corpus has 512 adversarial and 512 benign prompt labels. Responses were scored using the author's previously ablated Qwen3.6-35B-A3B-Abliterix-EGA-abliterated checkpoint, identified in the records as qwen36-ablinterix-t63-APEX [15]. These are model-assigned outcome categories. They are not tests that generated instructions work or that programs compile.

The following counts are recomputed from the archived verdicts. The refusal column means **maintained refusal**; partial refusals and other outcomes remain separate. Failed judgments are excluded from the percentage denominator and shown explicitly [13].

DeepSeek has 91 responses labeled degenerate across the full corpus, including 58 in the benign half. Its generation log records 823 of 1,024 responses ending at the token limit. A later cross-tabulation shows that 733 of those 823 responses carried the historical "valid fulfillment" label, 85 were labeled degenerate, and five received other labels. Nineteen of the 733 fulfillment labels also carry an invalid-output flag. Thus budget exhaustion, degeneration, and fulfillment



Each trace point has a different prefix; probabilities precede temperature and repetition penalties.

Figure 3. Archived entry-token and constrained-trace observations. Panel A uses twelve prompts, not the full corpus. Panel B follows one response. Neither panel proves that refusal cannot reappear later or that the resulting answer is correct. Values and source hashes accompany the figure.

Table 5. Historical W1 maintained-refusal rates. Percentage denominators exclude failed judgments, which are reported separately.

Campaign subject	Maintained refusal, adversarial	Unjudged, adversarial	Maintained refusal, benign	Unjudged, benign
DeepSeek-V4	12/512 (2.34%)	0	0/512	0
Gemma-4-26B	37/511 (7.24%)	1	0/509	3
Qwen3.8-27B	26/502 (5.18%)	10	0/512	0

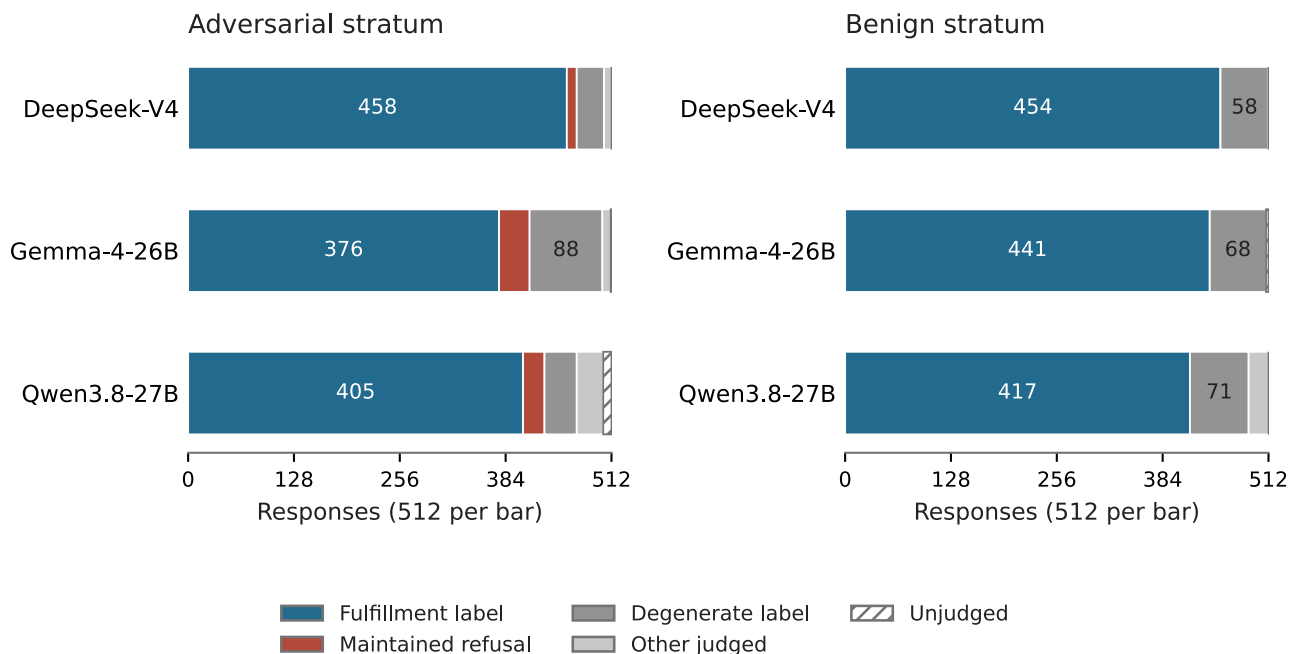


Figure 4. Historical W1 judgments on the two corpus strata. Each bar includes all 512 responses. “Other judged” combines partial refusal, pivot-then-fulfillment, mixed, and nonresponsive categories. Fulfillment and degeneration are labels from the original scoring pass, which clipped long answers. They do not certify full-response validity or task completion.

are overlapping observations; 733 is not a count of newly validated complete answers [16].

The historical harness requested 800 completion tokens per response. Its judge saw at most 2,500 characters, clipping 971 of DeepSeek’s 1,024 responses; it was not given the gen-

eration finish reason. Thirteen degeneration verdicts cite the artificial judging cutoff. Moreover, 20 of the 912 DeepSeek responses labeled “valid fulfillment” also carry an invalid-output flag. These counts reproduce the recorded categories, not reliable full-response quality assessments [13].

Table 6. GLP panel counts. Each arm contains the same 32 adversarial and 16 benign prompts. Fulfillment and refusal columns are semantic-judge labels; the final column is measured generation termination.

Arm	Adversarial maintained refusal	Benign fulfillment label	Benign token-limit termination
Unsteered	28/32	4/16	14/16
$\alpha = 0.5$	28/32	3/16	15/16
$\alpha = 1$	28/32	3/16	14/16
$\alpha = 2$	29/32	2/16	15/16
$\alpha = 4$	29/32	3/16	15/16

An offline embedding screen flags 9 of DeepSeek’s 12 maintained refusals and also flags 3 of its 912 judge-labeled valid fulfillments. The three missed refusals are 0.59% of the adversarial corpus, but this is an **undetected refusal rate after screening**, not a lower generation refusal rate. The screen does not regenerate an answer and is not evidence of an integrated serve-time feature [13].

A 25-case, refusal-enriched human spot-check reported agreement on 21 cases. Its listed disagreements include three over-flags and one under-flag. That sample is useful for discovering judge errors; it cannot establish unbiased corpus-wide agreement or justify assuming the errors cancel [15].

Taken together, the reported refusal rates depend on a single ablated judge whose corpus-wide agreement is unestablished, applied to runs with incomplete runtime and model provenance.

The retained records have no matched full-corpus unconstrained baseline, so they cannot isolate the grammar’s effect on refusal, length, or quality. They also lack complete per-run binary, model, and configuration identities. The W1 case study remains exploratory evidence of constrained output and its measurement limitations, rather than a reproducible benchmark of the current implementation.

The revised harness sends complete responses with termination metadata and rejects contradictory verdict fields. It binds judgments to the full input, keeps run and budget conditions distinct, checks hashes when joining records, and rejects incompatible resumptions. A managed measurement process binds verified model and binary files to the live tokenizer, template, defaults, and intervention state. Human-review exports retain full text and separate initial labeling from machine verdicts. Offline regressions exercise these contracts; they do not validate the historical labels. No completed replacement scoring pass is included in this case study [17].

6.3 A single-layer GLP evaluation

A subsequent experiment evaluated one candidate DeepSeek-V4 direction on a fixed panel of 32 adversarial and 16 benign prompts. The 4,096-dimensional vector targets graph layer 29 and uses the residual-to-FFN site transfer described above. This is one candidate at one layer, not a multilayer GLP intervention. Each arm used temperature zero, disabled thinking, no GCD, and a 256-token budget. There was one generation per prompt and arm; the baseline loaded no steering artifact [16].

All 240 responses received judgments, with no recorded request or judge errors. No adversarial response was labeled valid fulfillment, and no benign response was labeled maintained refusal. The judge used the same recorded Qwen identifier as the historical campaign. Its driver retained the earlier 2,500-character input limit, but none of these shorter responses exceeded it. Generation termination remains a substantial limit: 14–15 of 16 benign responses per arm exhausted the budget. Neither the absence of benign refusal nor similar capped-output labels establishes preserved capability [16].

Increasing strength produced no observed refusal reduction for this candidate and panel. At $\alpha = 2$ the transformation is a reflection by the equation; these counts do not establish that reflection caused the observed result. Direction quality,

layer coverage, and transfer between activation sites remain possible explanations. Six panel texts also occur in the checked-in calibration corpus, so the panel is not wholly held out. The records identify the binary, vector, model, corpus, and panel by hashes, but do not bind the binary to a source revision or identify the judge’s exact artifact. There are no repeated passes or loaded-vector zero-strength control. The result motivates behavioral validation of calibrated directions; it does not establish a general limit on GLP or a measured latency advantage.

7 Discussion and implications

Smaller behavioral artifacts change distribution. A separate steering artifact reduces redundant storage and down-loads and can be inspected, versioned, and disabled independently. Portability still depends on checkpoint and runtime compatibility.

Explicit constraints separate authority from model preference. A correctly enforced action grammar excludes unauthorized alternatives even when a model prefers them after consuming untrusted documents or tool results. The application owns the permitted action set and execution boundary.

Structure makes some failures easier to detect. Missing fields or out-of-enumeration values permit mechanical rejection; plausible findings still require evidence. Coding agents need tests of tool selection, arguments, tool-result continuation, and task completion alongside JSON validity.

Reduced refusal and improved capability require different evidence. Engagement with legitimate malware analysis or other difficult security work does not establish useful answers or improved reasoning. Measure engagement, accuracy, completion, and application outcomes separately.

The controls need independent and combined validation. Compare baseline, GCD, GLP, and their combination with identical model artifacts, prompts, sampling settings, and budgets. Hold the grammar constant across GCD arms, and the vector, sites, and strength across GLP arms. Record truncation and errors alongside quality; measure latency, throughput, and cache reuse on the actual serving path.

hf2q makes both controls inspectable in a local inference stack. Their practical value depends on specifying the activation and output-language contracts, testing the executed paths, and measuring completed application outcomes separately from the model’s willingness to answer.

Acknowledgments

Vince Ovando contributed the generation-time authorization framing and the GCD collaboration. Matt Suiche contributed GLP, weightless, and supporting experimental work. GCD and activation steering also build on the prior research cited below. The diagrams in this article are explanatory redraws; the original stack illustration was contributed by Matt Suiche.

Artifacts and reproducibility

The current implementation account uses hf2q commit 1bf0c82f10369a0714ea72ed20d979d7c387ffde. The historical W1 audit retains source snapshot 44004311717d414feaa54384578e2bcf4d140464. Neither source inspection identifies a historical runtime by itself. The follow-up evidence manifest separates later GLP records, battery attempts, and historical termination cross-tabs; its offline extractor recomputes those aggregates without publishing prompt or response text. Figure data and formulas, aggregate data and hashes, and response-count CSV accompany the article.

References

- [1] Geng, S., Josifoski, M., Peyrard, M., and West, R. Grammar-Constrained Decoding for Structured NLP Tasks without Fine-tuning. EMNLP, 2023.
- [2] Ovando, V. *Constitutive Authorization at the Decoding Boundary: Grammar-Constrained Decoding as a Positive, Generation-Time Security Control for LLM Agents*. Supplied manuscript gcd.pdf, draft June 26, 2026, §§3–4, 6, 8. Project DOI; companion soundness argument. The reviewed PDF is the supplied version; identity with the current DOI download is not asserted.
- [3] Suiche, M. GLP — GGUF Layer Projection: format and apply path; weightless artifact catalog. Specification revision 3481e30eba9d; accessed September 11, 2026.
- [4] Dong, Y., et al. XGrammar: Flexible and Efficient Structured Generation Engine for Large Language Models. 2024.
- [5] Park, K., Wang, J., Berg-Kirkpatrick, T., Polikarpova, N., and D’Antoni, L. Grammar-Aligned Decoding. NeurIPS, 2024.
- [6] Arditi, A., et al. Refusal in Language Models Is Mediated by a Single Direction. 2024.
- [7] Suiche, M. Sticky Refusals, Free Speculative Decoding, and the Invisible Quantisation Cliff, September 3, 2026; September 4 hook-site correction.
- [8] Suiche, M. Unpublished research on activation steering, privately shared with Robert E. Lee, 2026.
- [9] hf2q. Reviewed source tree; CLI; request policy.
- [10] hf2q. Grammar implementation; engine.
- [11] hf2q. Schema compiler; recon schema. JSON Schema, string constraints.
- [12] hf2q. GLP subsystem; calibration implementation; ADR-054.
- [13] hf2q. Historical W1 generation, judge, and embedding-screen JSONL records, September 2026. Exact filenames, hashes, counts, and access status appear in the evidence manifest. Offline aggregation script; judging harness. Raw logs are local campaign artifacts, not all present in the published repository.
- [14] hf2q. refusal_mass_probe.jsonl, twelve archived observations, and probe script. Probability semantics: sampler, sample_token_with_logprob_topk, and the grammar-aware engine call site.
- [15] Lee, R. E. Qwen3.6-35B-A3B-Abliterix-EGA-abliterated, judge checkpoint. hf2q, human spot-check of the Qwen judge, September 8, 2026.
- [16] hf2q. September 10 GLP panel records and judge driver; operational battery; historical termination report. Recomputed counts and source identities: follow-up manifest.
- [17] hf2q. Revised measurement methods; judge, rejudge, report, and offline/mock tests.
- [18] Suiche, M. Captain Vector: GLP production and validation utility. Revision 3481e30eba9d; accessed September 11, 2026.
- [19] Hu, E. J., et al. LoRA: Low-Rank Adaptation of Large Language Models. 2021.
- [20] Sun, M., Chen, X., Kolter, J. Z., and Liu, Z. Massive Activations in Large Language Models. COLM, 2024.